LifeLink, Blood Donation Matching Platform Diagrams

Group 3

Diagram 1 Use Case Diagram

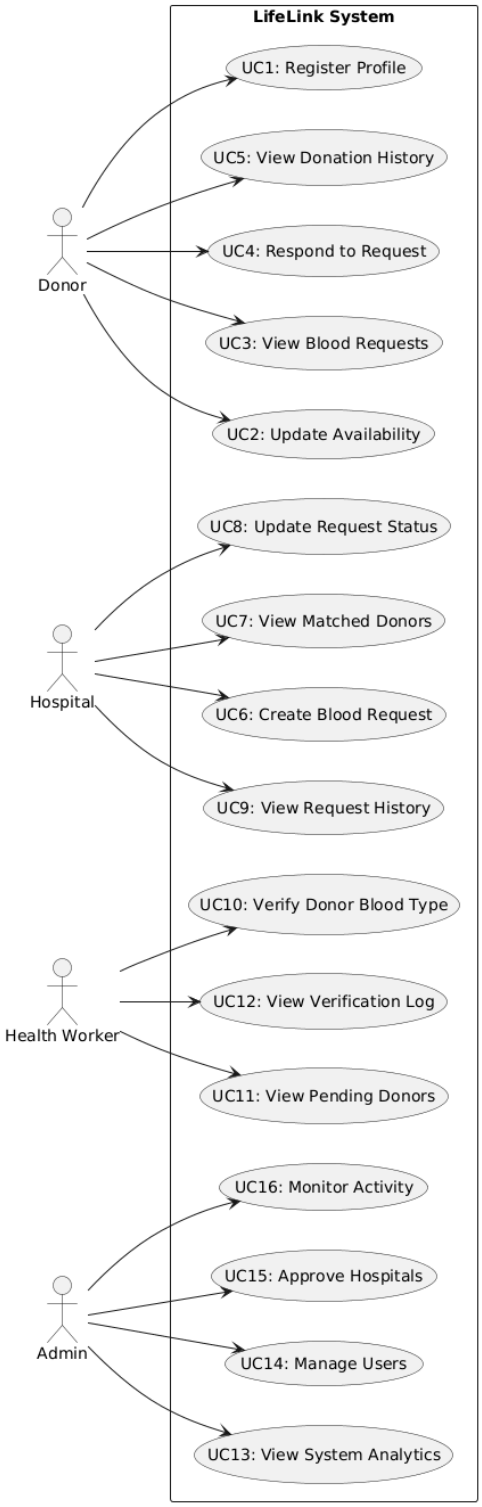

Diagram 2 Class Diagram

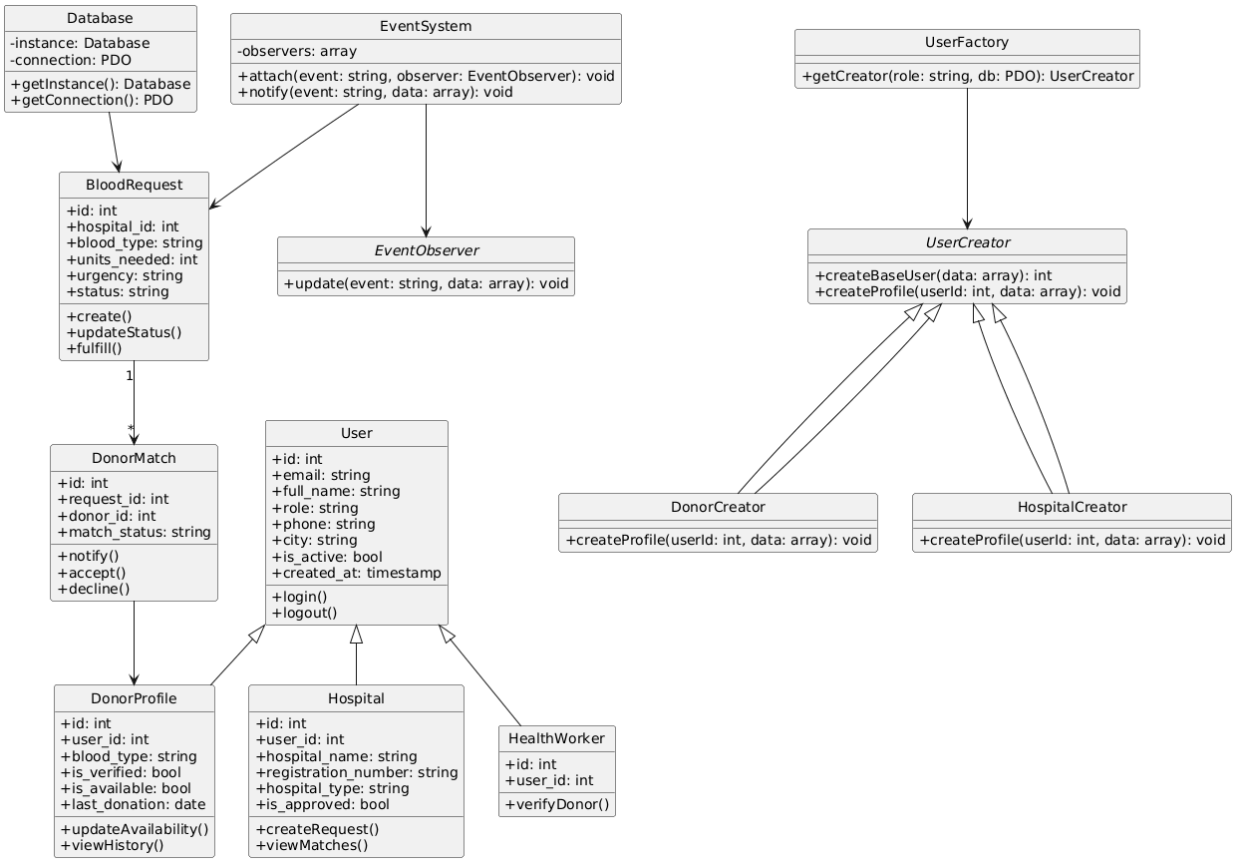

Diagram 3 Activity Diagram: Blood Request Flow

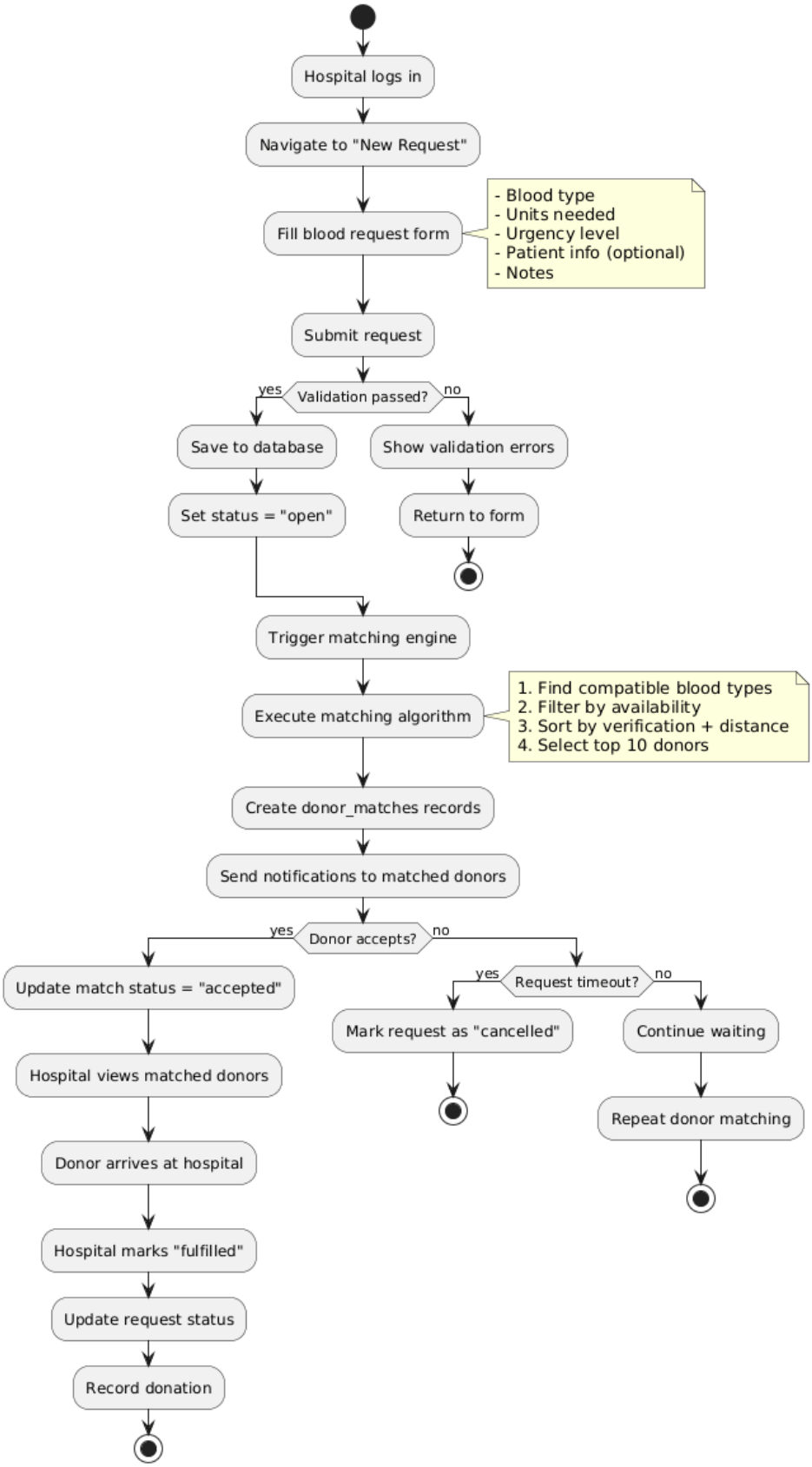
Hospital logs in
Navigate to "New Request"
Fill blood request form
- Blood type
- Units needed
- Urgency level
- Patient info (optional)
- Notes
Submit request
Validation passed?
yes
no
Save to database
Show validation errors
Set status = "open"
Return to form
Trigger matching engine
Execute matching algorithm
1. Find compatible blood types
2. Filter by availability
3. Sort by verification + distance
4. Select top 10 donors
Create donor_matches records
Send notifications to matched donors
Donor accepts?
yes
no
Update match status = "accepted"
Request timeout?
yes
no
Mark request as "cancelled"
Continue waiting
Hospital views matched donors
Repeat donor matching
Donor arrives at hospital
Hospital marks "fulfilled"
Update request status
Record donation

Diagram 4 Activity Diagram: Donor Registration Flow

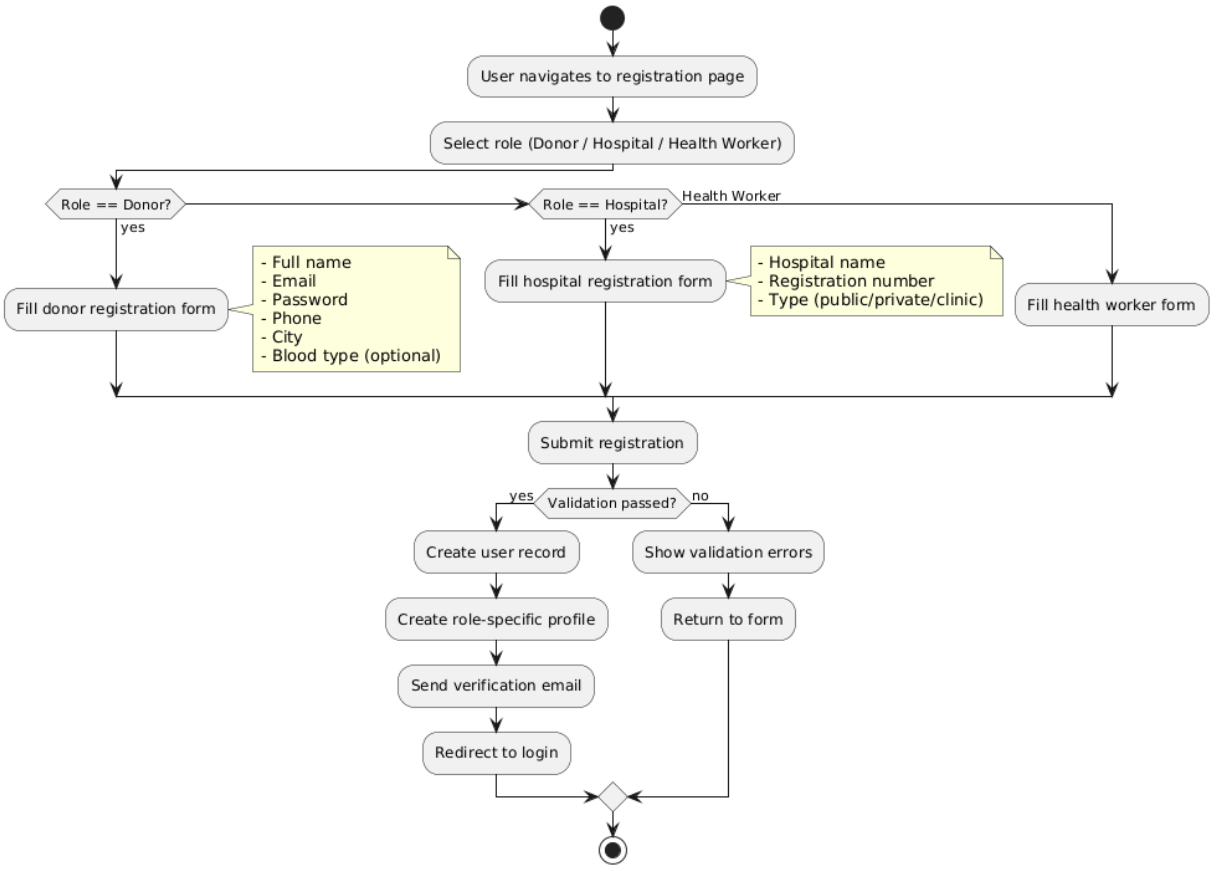
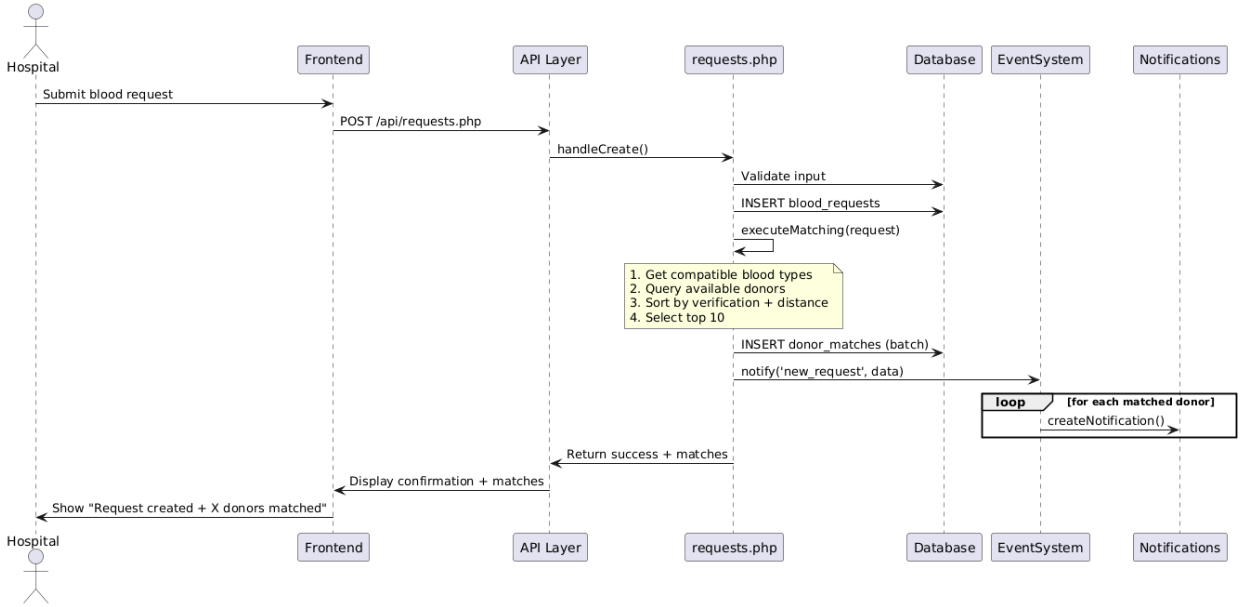
Diagram 5 Sequence Diagram: Matching Engine